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1. Introduction to Web Development

What is Web Development?

Web Development is the process of **creating websites or web applications that run on the internet**.

It includes:

1. **Frontend Development** – What users see
2. **Backend Development** – Server logic and database
3. **Database** – Where data is stored

   Example: Amazon Website

Component	Technology	Work
Frontend	HTML, CSS, JavaScript	Shows products
Backend	Node.js	Handles orders
Database	MongoDB / MySQL	Stores users & products

2. Types of Web Development

1. Frontend Development

Frontend means **user interface (UI)** of the website.

Technologies used:

- HTML
- CSS
- JavaScript
- Angular / React / Vue

Example

Login page, buttons, forms.

```
<!DOCTYPE html>
<html>
<head>
<title>Login Page</title>
</head>

<body>

<h2>Login Form</h2>

<form>
Username: <input type="text"><br><br>
Password: <input type="password"><br><br>

<button>Login</button>
</form>

</body>
</html>
```

2. Backend Development

Backend handles:

- Server logic

- Authentication
- Data processing

Technologies:

- Node.js
- PHP
- Python
- Java

Example:

User login validation.

```
app.post('/login', (req,res)=>{
  const username = req.body.username;
  const password = req.body.password;

  if(username == "admin" && password=="1234"){
    res.send("Login Success");
  }else{
    res.send("Invalid user");
  }
});
```

3. Database

Database stores information.

Example data:

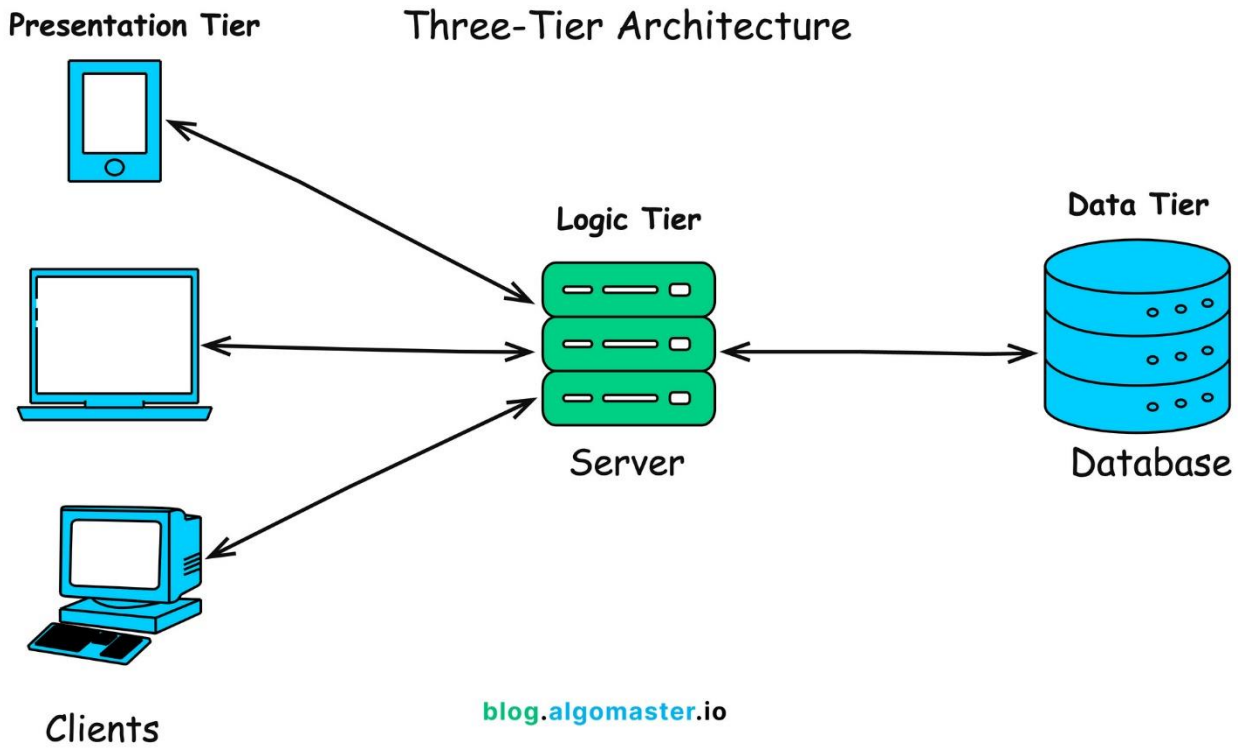
Id	name	email
1	Ashwani	Ash123@gmail.com
2	Jadu	Jadu141@gmail.com

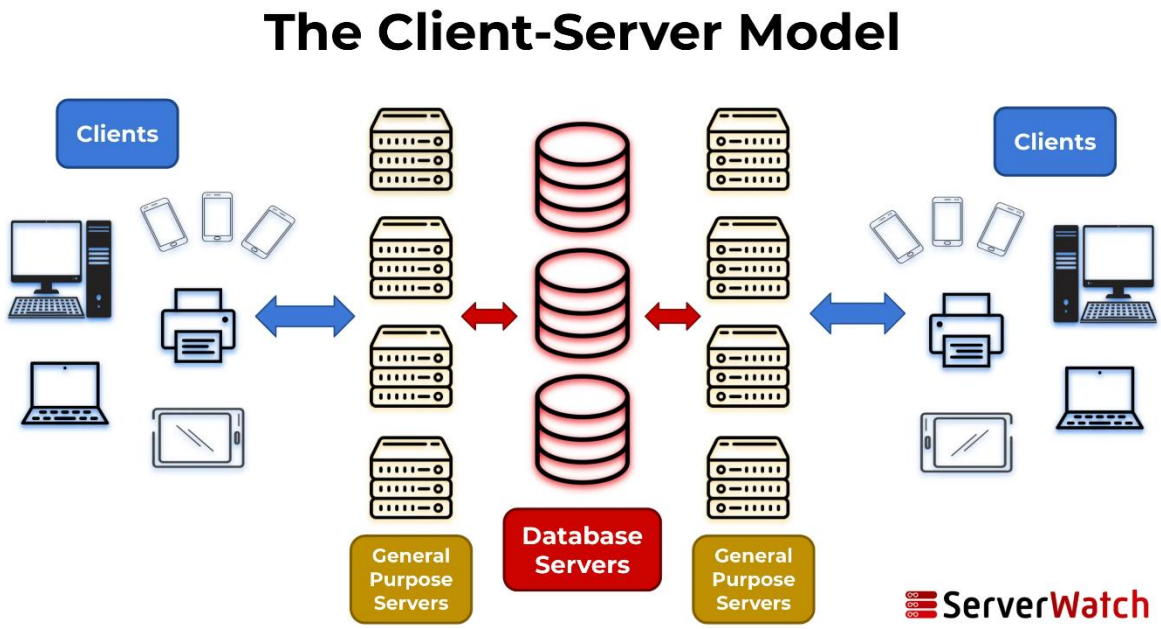
Databases:

- MongoDB
- MySQL
- PostgreSQL

Web Application Architecture

Client Server Architecture





Explanation:

1. **Client (Browser)**
   User opens website in Chrome.
2. **Server**
   Processes request.
3. **Database**
   Stores data.

Flow:

User → Browser → Server → Database

Example:

Student opens **college website**

1 Request goes to server
2 Server fetches student data
3 Server sends response to browser

3. Introduction to MEAN Stack

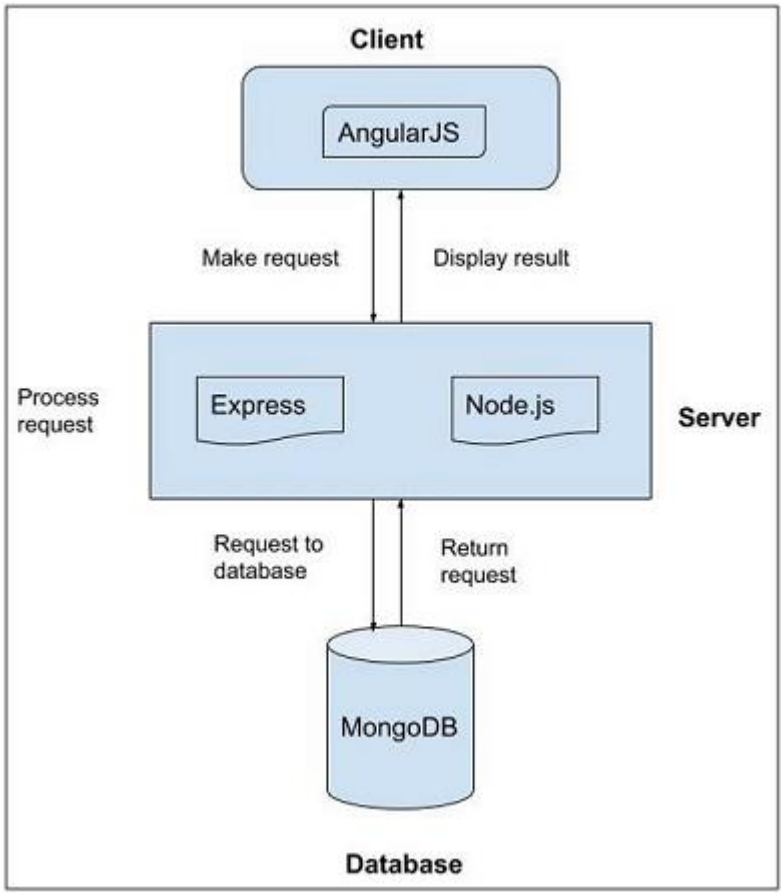
What is MEAN Stack?

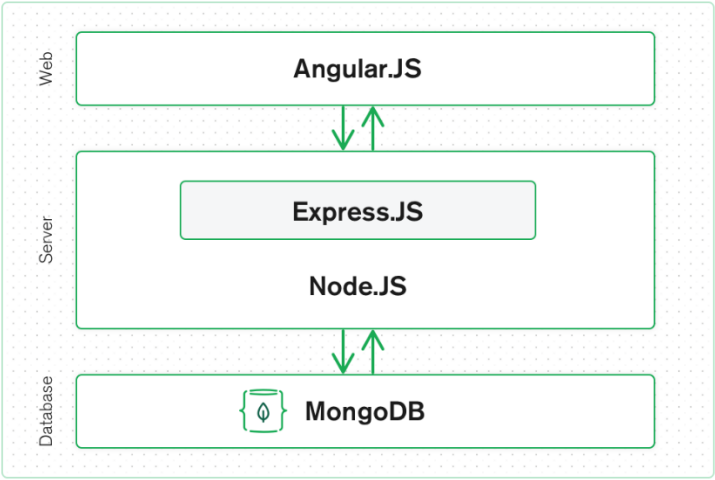
MEAN Stack is a JavaScript based full stack development framework.

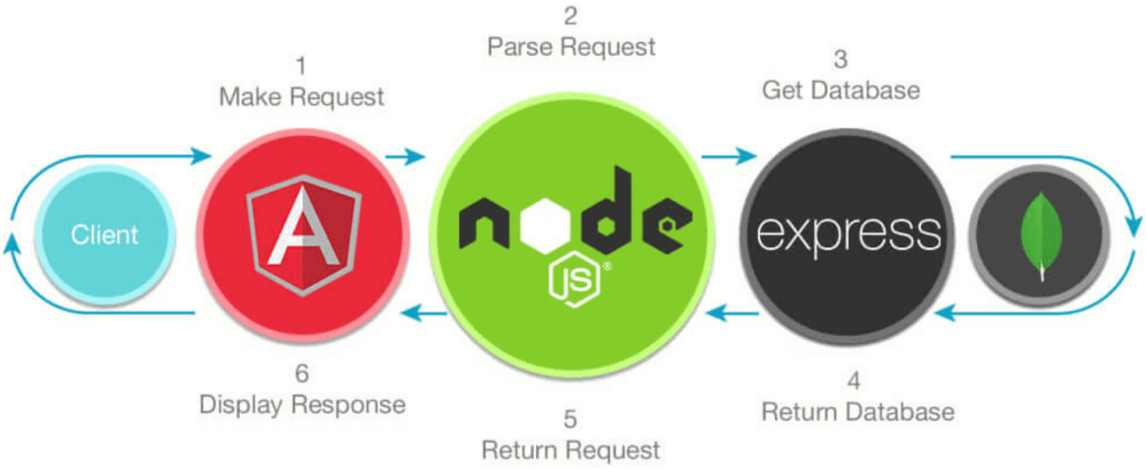
MEAN stands for:

Letter	Technology	Work
M	MongoDB	Database
E	Express.js	Backend Framework
A	Angular	Frontend Framework
N	Node.js	Server Runtime

MEAN Stack Architecture




Web
Angular.JS
Server
Express.JS
Node.JS
Database
MongoDB


1
Make Request
2
Parse Request
3
Get Database
Client
express
6
Display Response
5
Return Request
4
Return Database

Flow:

Angular (Frontend)

↓

Node.js + Express (Server)

↓

MongoDB (Database)

Example

Student Registration System

Layer	Work
Angular	Student registration form
Node + Express	API for saving student
MongoDB	Store student data

4. Components of MEAN Stack

1. MongoDB

- MongoDB is a **NoSQL database**.
- It stores data in **JSON format**.

Example Data

```
{
  name: "Rahul",
  course: "BCA",
  age: 21
}
```

Advantages:

- Fast
- Flexible
- JSON based

2 Express.js

- Express is a backend framework for Node.js.
- It helps create APIs easily.

Example API

```
const express = require('express')

const app = express()

app.get('/', (req,res)=>{

  res.send("Welcome to MEAN Stack")

})

app.listen(3000)
```

Output:

http://localhost:3000

3 Angular

Angular is a frontend framework developed by Google.

Used for:

- Single Page Applications
- Dynamic UI

Example Angular Component

```
import { Component } from '@angular/core';

@Component({

 selector: 'app-root',

 template: '<h1>Hello Students</h1>'

})

export class AppComponent {}
```

Output:

Hello Students

4 Node.js

- Node.js is a JavaScript runtime environment.
- It allows JavaScript to run on server.

Example:

```
const http = require('http')

http.createServer((req,res)=>{

  res.write("Hello from NodeJS")

  res.end()

}).listen(3000)
```

Output:

http://localhost:3000

5. Advantages of MEAN Stack

- 1 Single language JavaScript everywhere
- 2 Fast development
- 3 Large community support
- 4 JSON based data transfer
- 5 Scalable applications

Example companies using MEAN:

- Netflix
- PayPal
- LinkedIn

6. Setting Up the Development Environment

Step 1 Install Node.js

Download from

https://nodejs.org

Check installation

node -v

Example output

v18.0.0(Latest Version)

Check npm:

npm -v

Step 2 Install Angular CLI

Command:

npm install -g @angular/cli

Check installation

ng version

Step 3 Install MongoDB

Download:

https://www.mongodb.com

Start MongoDB server

Mongod

Open MongoDB shell

Mongo

Step 4 Install Express

Create project

mkdir mean-project
cd mean-project

Initialize project

npm init -y

Install Express

npm install express

Example Simple Express Server

Create file:

server.js

Code:

```
const express = require('express')

const app = express()

app.get('/', (req,res)=>{

   res.send("MEAN Stack Server Running")

})

app.listen(3000, ()=>{

  console.log("Server started on port 3000")

})
```

Run:

node server.js

Open browser:

http://localhost:3000

Output:

MEAN Stack Server Running